Lab 3: Architecture Justification

Problem Statement #14: Corporate Mental Wellness & Counseling Platform · Vivek Varma · PES1UG24AM070 · Section 5B

Architecture selection: *We chose Microservices Architecture (with an event bus) for the Corporate Mental Wellness & Counseling Platform.*

Architectural choice

Each actor gets its own portal (Employee, Therapist, HR Admin), and all three call one API Gateway. Behind it, five domain services each own a private database: Assessment & Triage, Availability, Booking, Reporting and Identity. Services talk synchronously only where an answer is needed at once (the booking slot lock). Everything else goes through an event bus, for example a severe score triggering a priority slot, or a booking triggering a notification. Layered was rejected because its shared data tier would place employee identities next to clinical data. Client-server was rejected because one central server cannot scale booking peaks on its own and is a single point of failure.

Reason 1: Database per service keeps employees anonymous

FR-001 and NFR-002 require that no counselling record can be traced back to an employee. Here the employee ↔ pseudo-ID mapping exists only in the Identity Vault, reachable only through the Identity Service. Clinical DB, Booking DB and Report Store contain pseudo-IDs alone. The separation is enforced by deployment and credentials, not just by coding discipline.

Reason 2: Booking has very different load and correctness needs

Booking must never double-book (FR-003) and must return slots in under 2 seconds for 500 concurrent users across timezones (NFR-001). Other parts of the system, such as monthly HR reports, are light and infrequent. Splitting Availability and Booking into their own services lets the team tune and scale that path alone. The atomic slot hold (ISlotLock) stays inside one small, well-tested boundary.

Security advantage

Compromising one service does not compromise the platform. The gateway’s auth filter accepts only TLS 1.3 traffic carrying a pseudo-ID JWT. Each database uses its own AES-256 key and credentials. HR’s console can reach only the Reporting Service, which reads anonymised events and refuses any group smaller than ten. HR therefore has no technical path to an individual’s data.

Performance benefit

The event bus removes slow work from the user’s request. A booking commits its slot and publishes BookingConfirmed straight away. Confirmations, reminders and report updates are then handled asynchronously by the Notification Worker and the Reporting Service. A slow email gateway or a heavy report therefore never delays booking, and Booking and Availability can be replicated horizontally at peak times without scaling anything else.